

Lecture 6: Thompson Sampling for Bandits

ID 6002W: Online and Reinforcement Learning

Web-enabled M.Tech. program, IIT Madras

General Principles

For multi-armed bandits

- Minimising regret involves careful exploration-exploitation tradeoff
- Algorithms should use data to calibrate uncertainty about arms to make decisions

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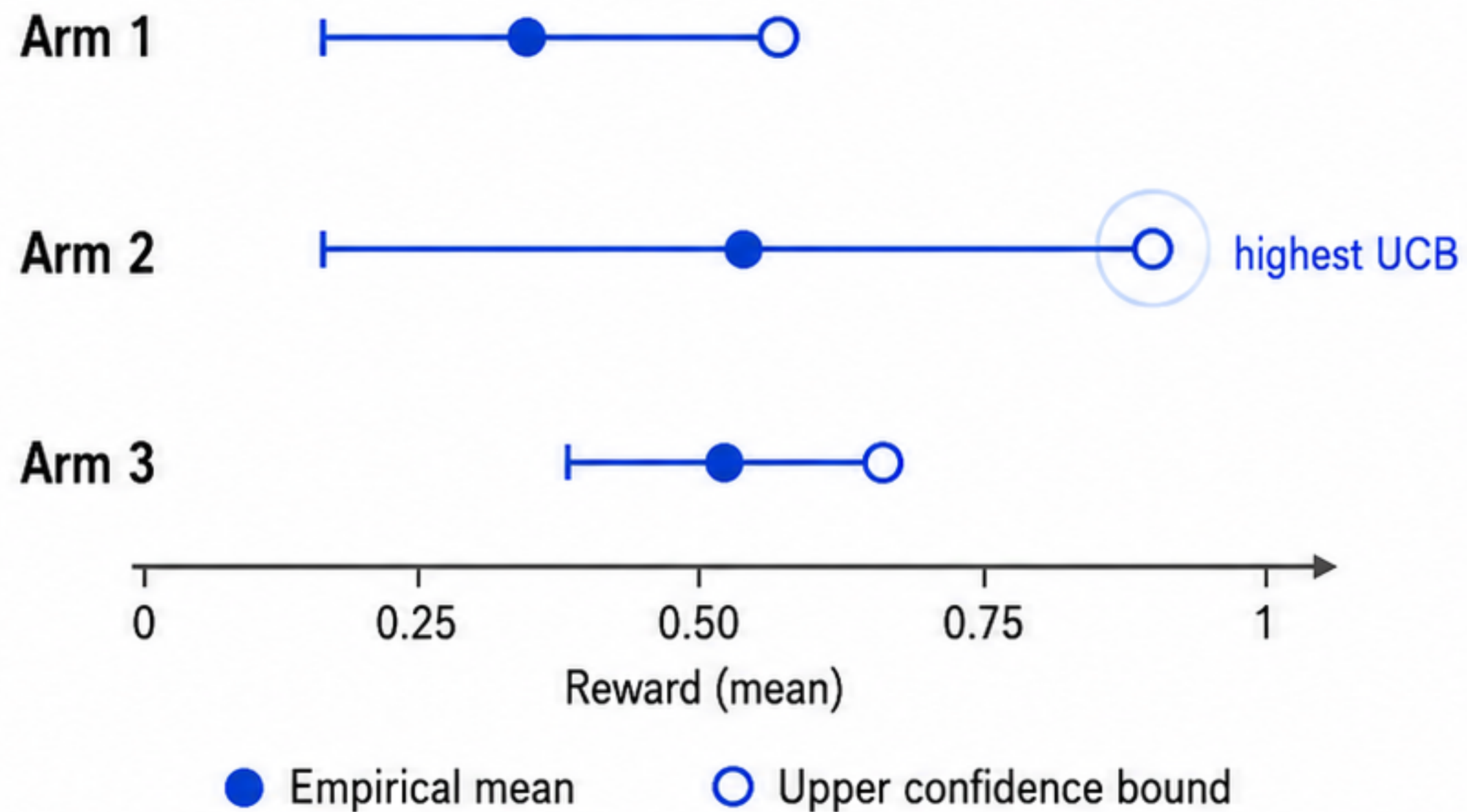
For multi-armed bandits

- Minimising regret involves careful exploration-exploitation tradeoff
- Algorithms should use data to calibrate uncertainty about arms to make decisions
- UCB and Thompson Sampling differ in how they represent and use uncertainty:
 - UCB maintains *confidence intervals*
 - Thompson Sampling uses *belief distributions*

UCB vs Thompson Sampling

UCB

Choose the highest upper confidence bound

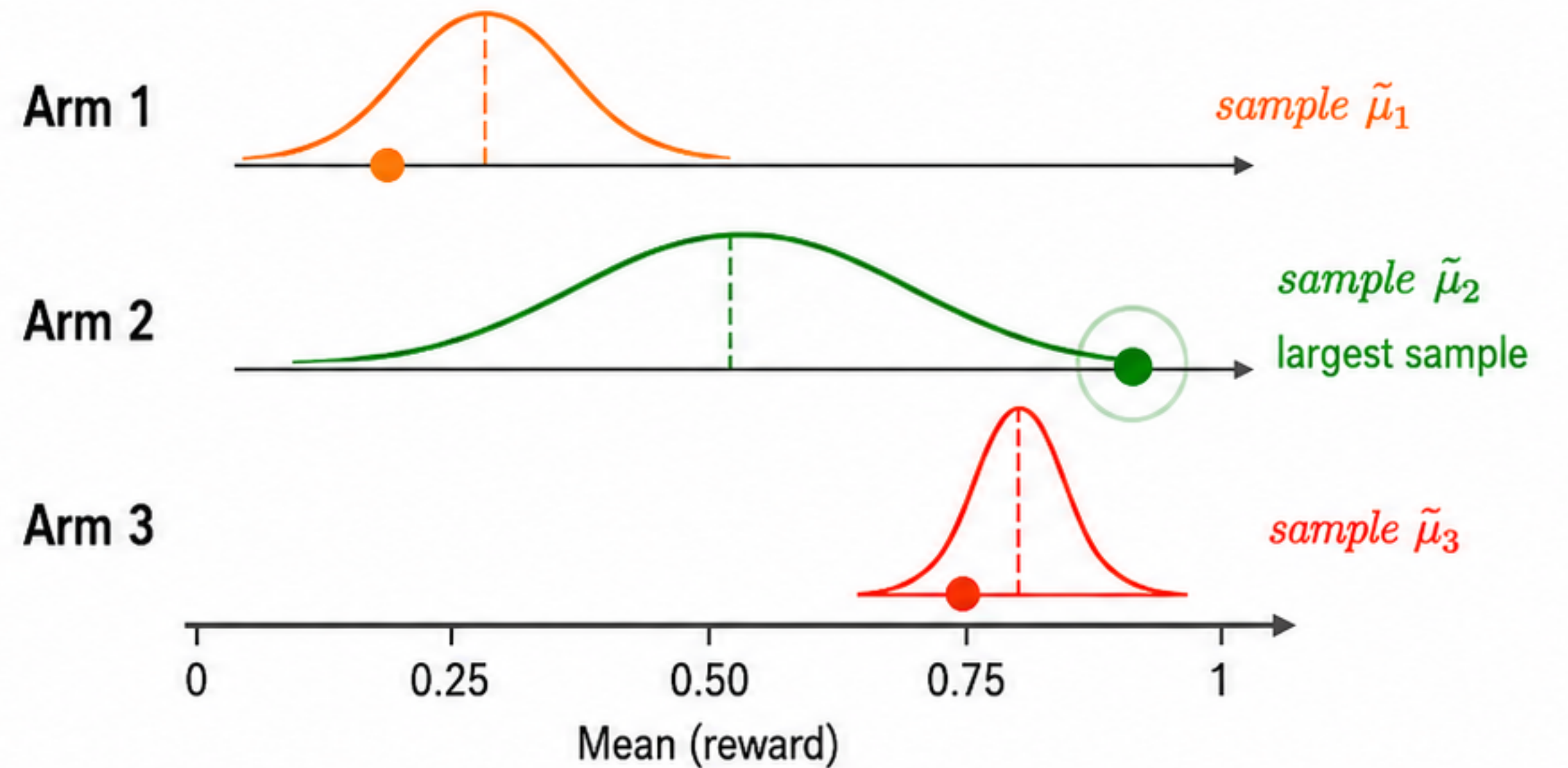


$$UCB_i(t) = \hat{\mu}_i(t) + bonus_i(t)$$

Deterministic: pick the arm with the largest optimistic value.

Thompson Sampling

Sample one plausible mean for each arm



$$\text{sample } \tilde{\mu}_i(t) \text{ for each arm, } A_t = \arg \max_i \tilde{\mu}_i(t)$$

Randomized: pick the arm with the largest sampled plausible value.

Bayesian Viewpoint

For multi-armed bandits

The true arm mean is *fixed but unknown*:

$$\mu_i = \mathbb{P}(X = 1 \mid \text{arm } i)$$

- ▶ With finite amount of i.i.d. data, we will always have some uncertainty about μ_i
- ▶ A belief distribution represents our uncertainty about μ_i

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- ▶ A belief distribution represents our uncertainty about μ_i

Data does not change μ_i — data changes our *belief* about μ_i

With more data, the belief usually gets more *concentrated*

Bayes' Rule

For updating beliefs

$$p(\mu \mid \text{data}) = \frac{p(\mu) \mathbb{P}(\text{data} \mid \mu)}{\mathbb{P}(\text{data})}$$

For simplicity, we write

$$p(\mu \mid \text{data}) \propto p(\mu) \mathbb{P}(\text{data} \mid \mu)$$

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- $p(\mu)$: Prior belief
- $p(\mu \mid \text{data})$: Posterior belief
- $\mathbb{P}(\text{data} \mid \mu)$: Likelihood

Prior $\times$ Likelihood $\rightarrow$ Posterior

Bayes' Rule

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We can also write:

$$p(\mu \mid \text{all data}) = p(\mu \mid \text{new data, old data}) \propto p(\mu \mid \text{old data}) \mathbb{P}(\text{new data} \mid \mu, \text{old data})$$

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- $p(\mu \mid \text{old data})$: Prior belief
- $p(\mu \mid \text{all data})$: Posterior belief
- $\mathbb{P}(\text{new data} \mid \mu, \text{old data}) = \mathbb{P}(\text{new data} \mid \mu) = \text{likelihood}$

Posterior of one round becomes
prior of the next round

i.i.d. data

Generating Samples from Beliefs

If we maintain a belief distribution $f_{i,t}(\mu)$ for each arm i at all times t ,

We can draw a random sample $\tilde{\mu}_i(t) \sim f_{i,t}(\mu)$

- Sample is drawn through simulation on a computer (easy to do)

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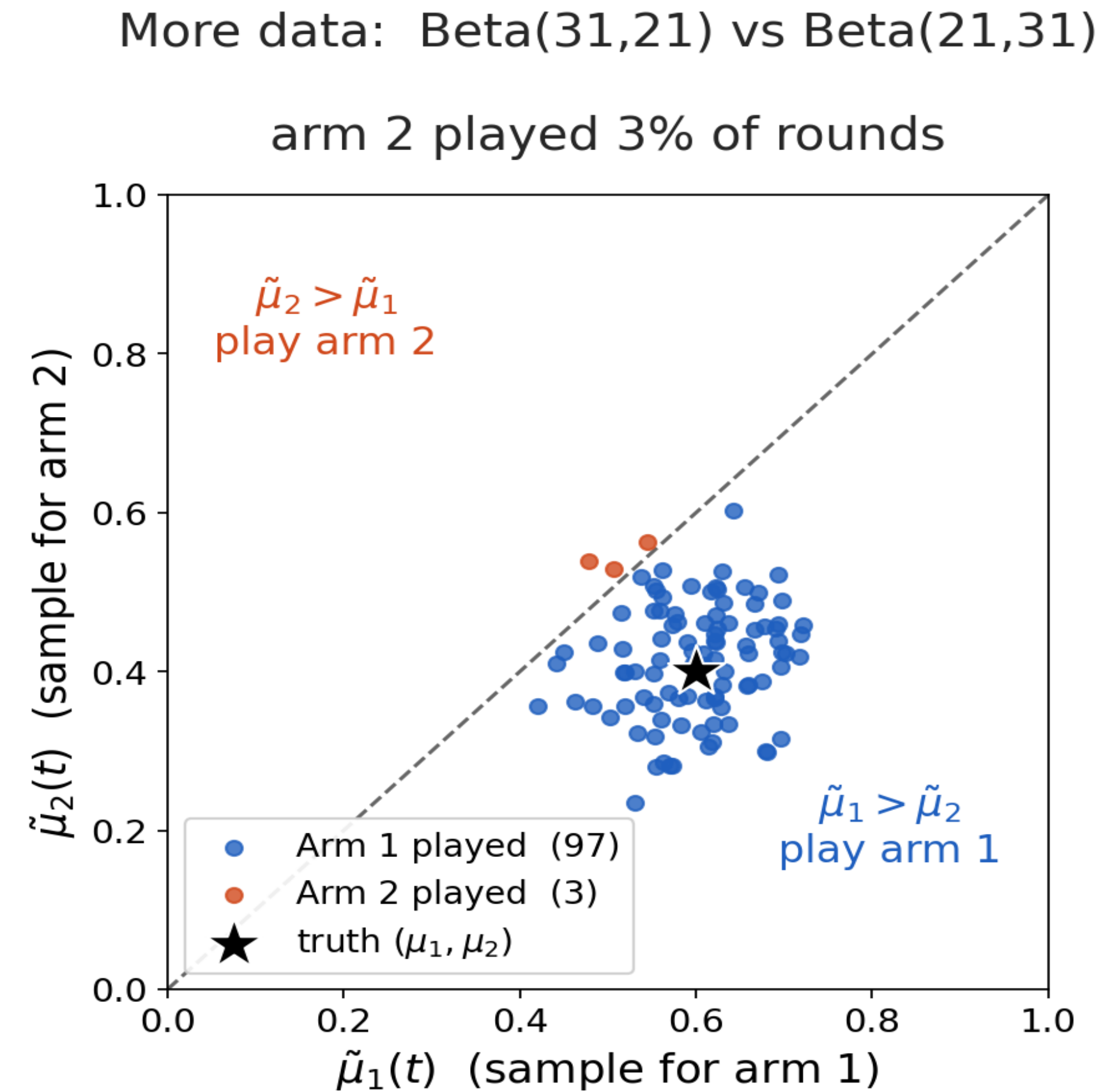
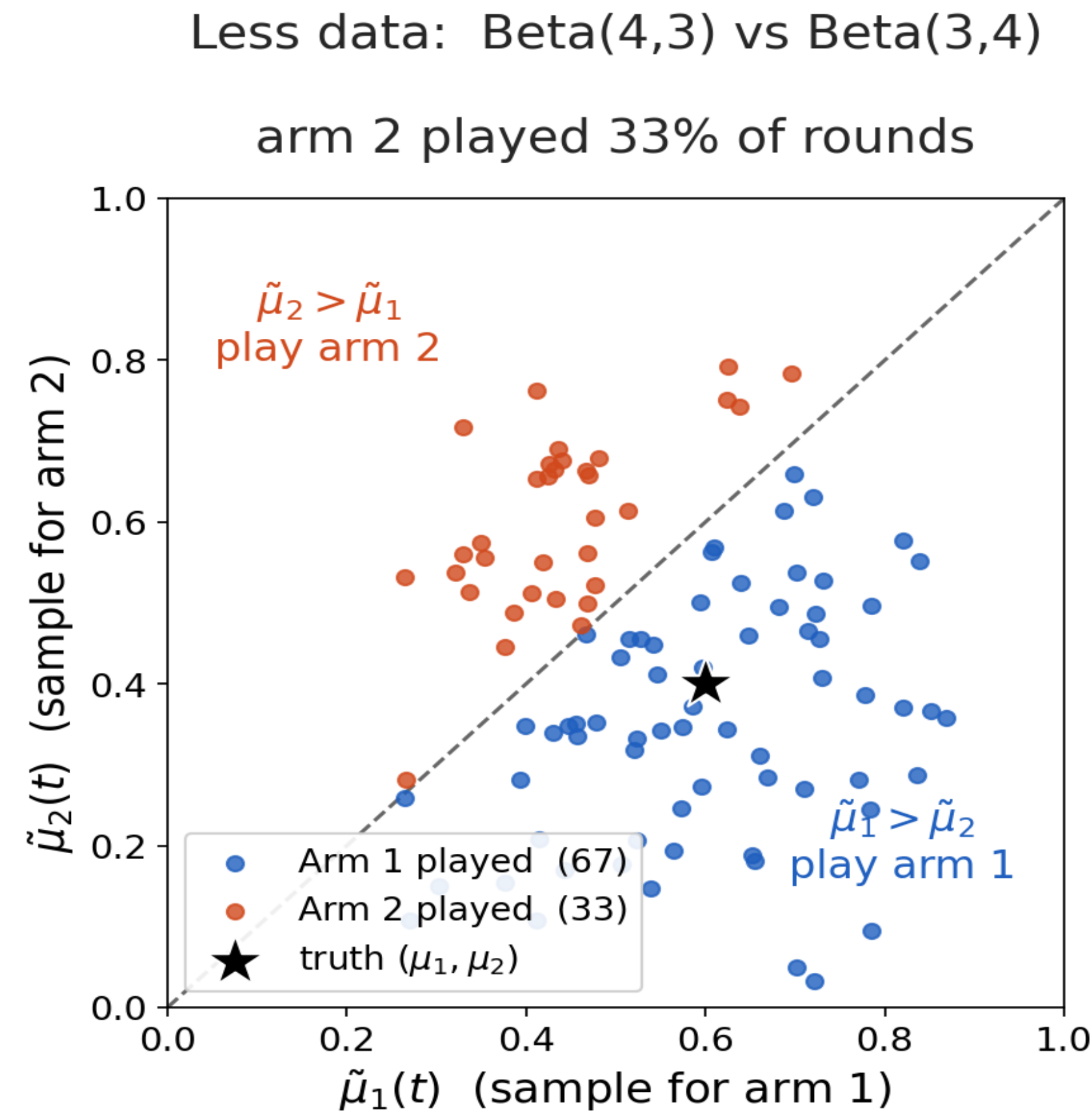
The sample $\tilde{\mu}_i(t)$ is a plausible value of μ_i (*not an empirical average*)

The set of samples $(\tilde{\mu}_1(t), \dots, \tilde{\mu}_K(t))$ are a set of plausible values of $(\mu_1, \dots, \mu_K)$

Thompson sampling rule: play as if the sampled world is the true world

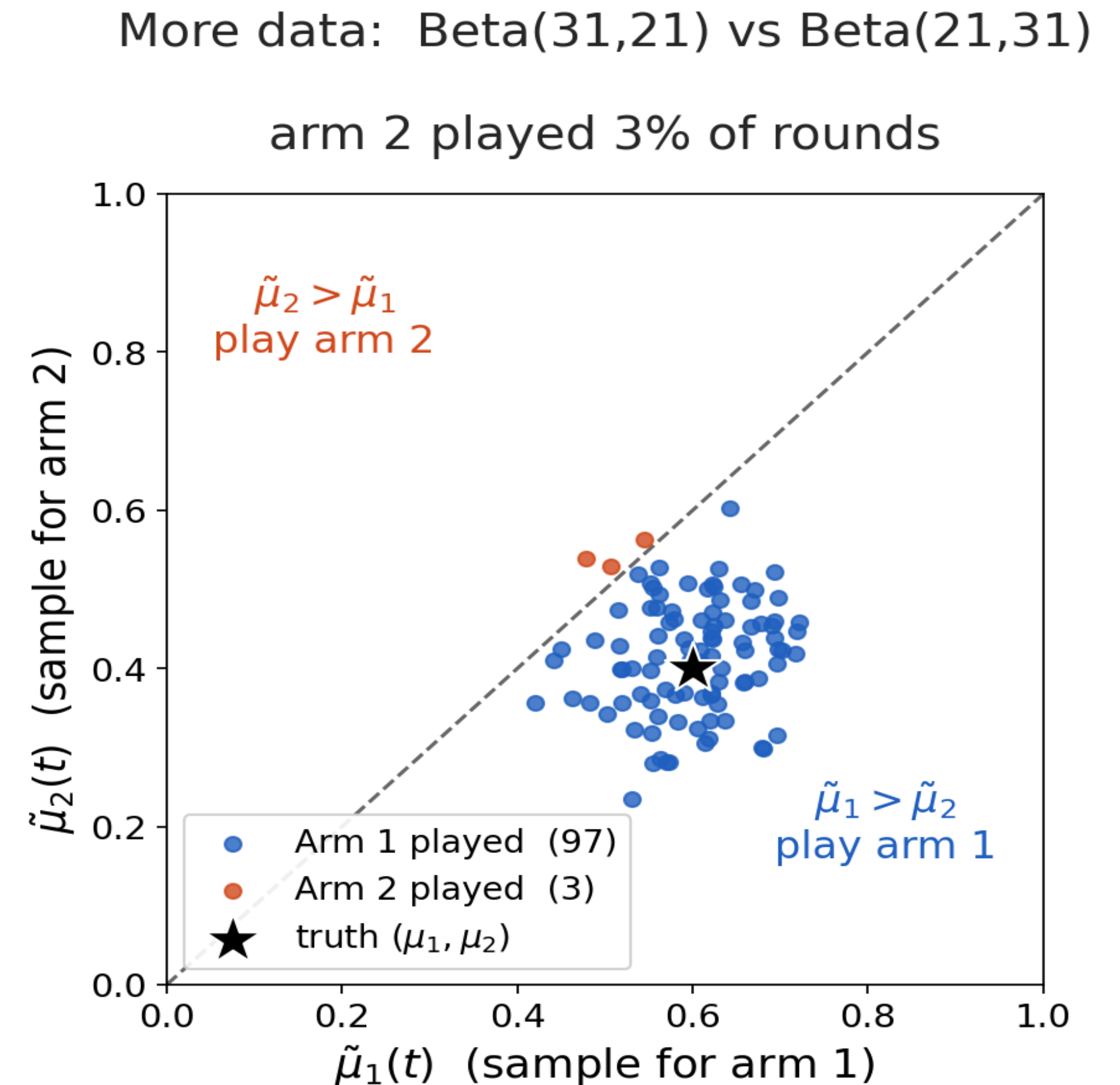
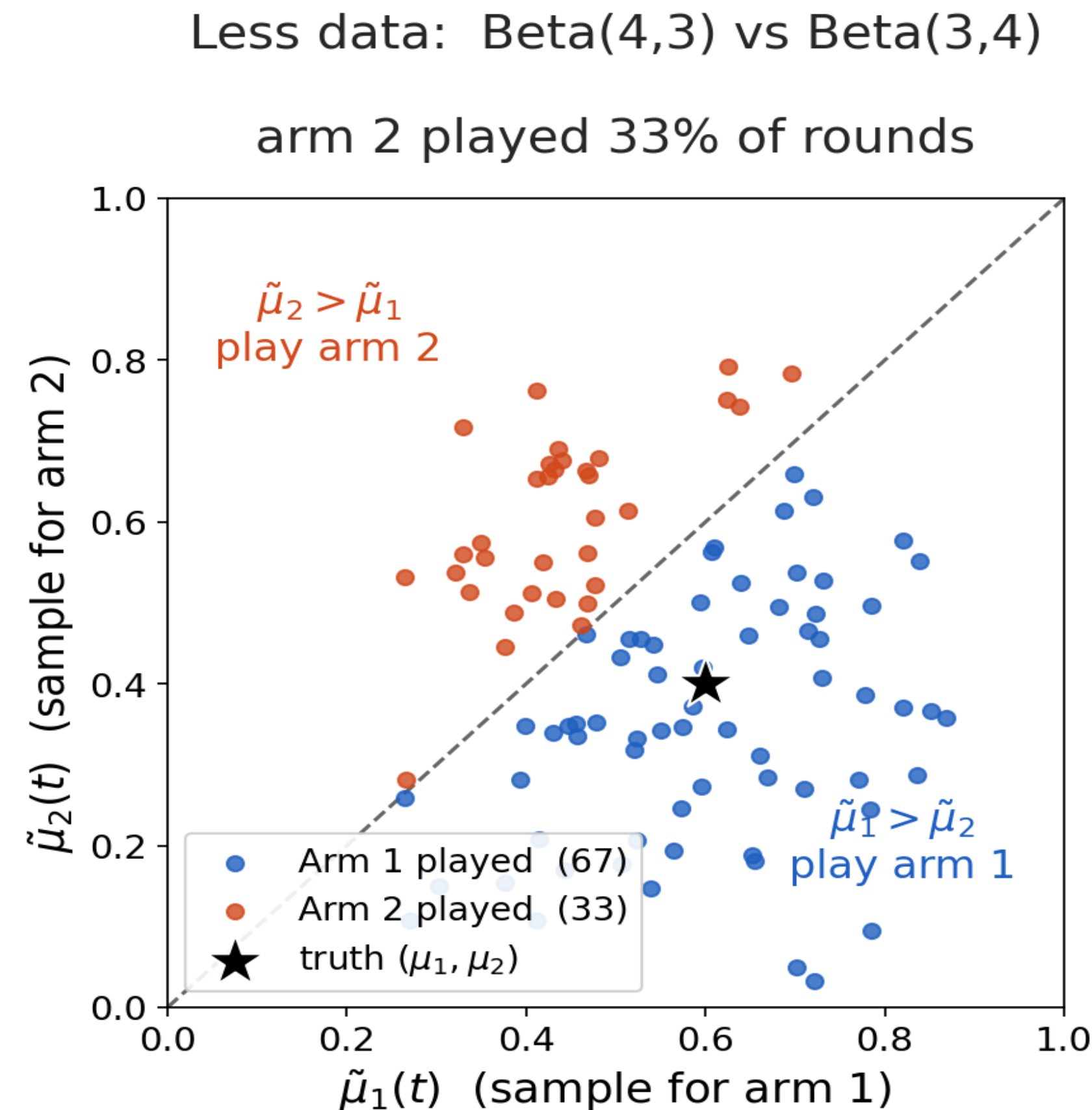
Why Thompson Sampling?

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Thompson sampling rule: play as if the sampled world is the true world

Leans towards exploration with less data, leans towards exploitation with more data

Making This Concrete: Bernoulli Bandits

What Do We Know About an Arm?

Bernoulli reward setting

For arm i , we can keep track of:

$S_i(t)$ = number of successes (ones)

$F_i(t)$ = number of failures (zeros)

Empirical average:

$$\hat{\mu}_i(t) = \frac{S_i(t)}{S_i(t) + F_i(t)}$$

How can we construct a belief distribution from $(S_i(t), F_i(t))$?

The Beta Distribution

A belief over mean rewards

For arms with Bernoulli rewards, $\mu_i \in [0,1]$

$\Rightarrow$ the belief distribution of μ_i should also take values in $[0,1]$

We use

$$\mu_i \sim \text{Beta}(\alpha_i, \beta_i)$$

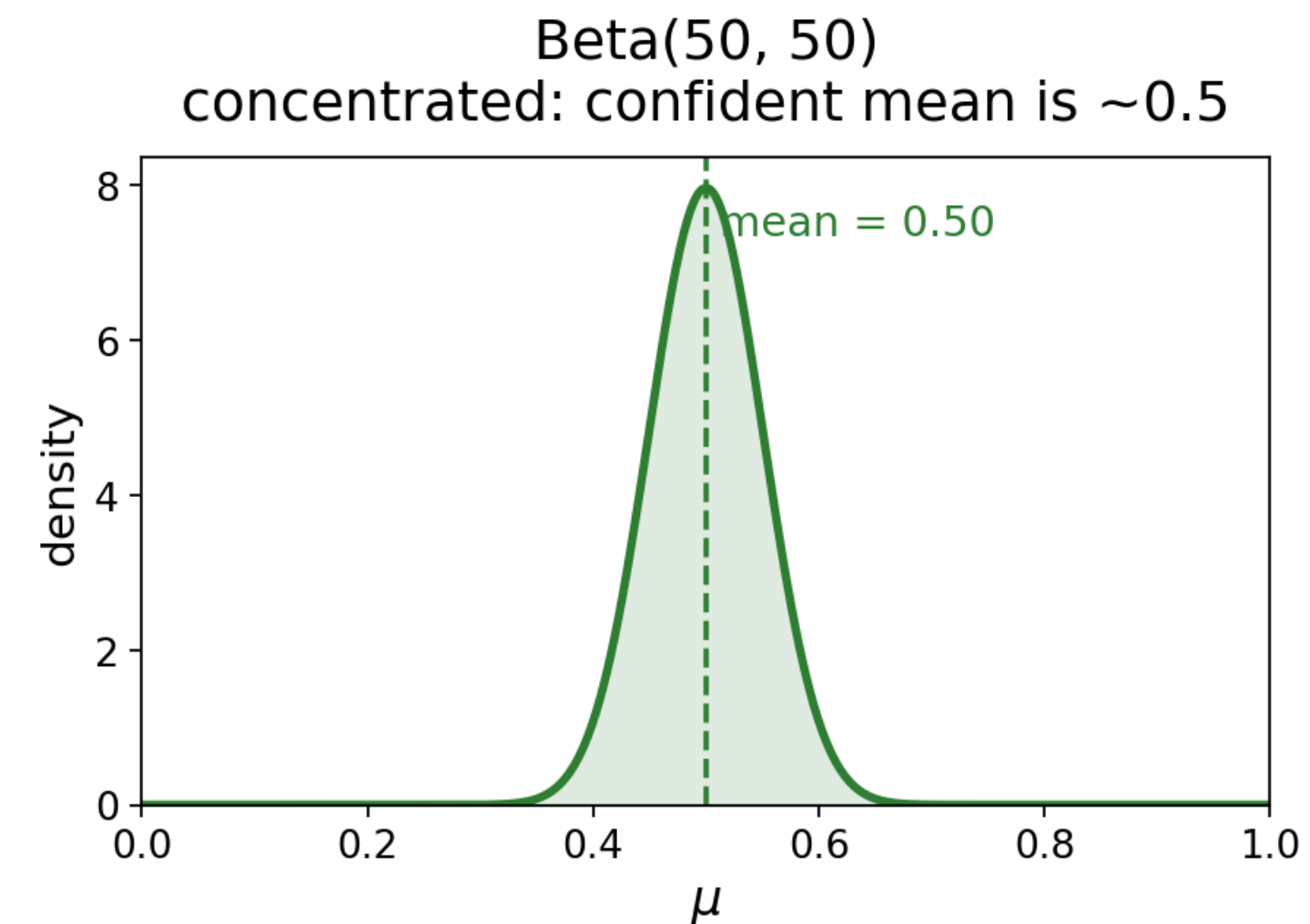
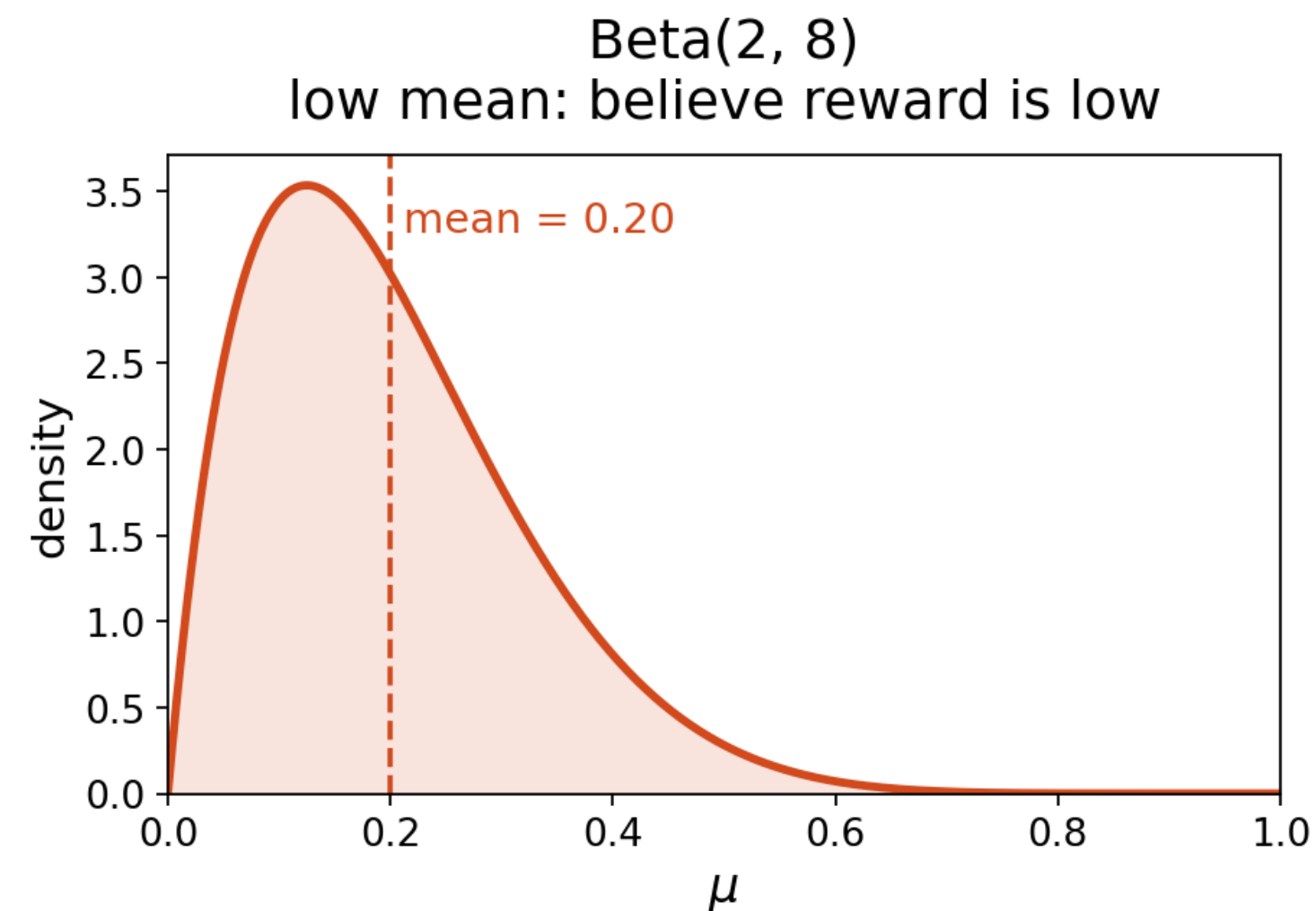
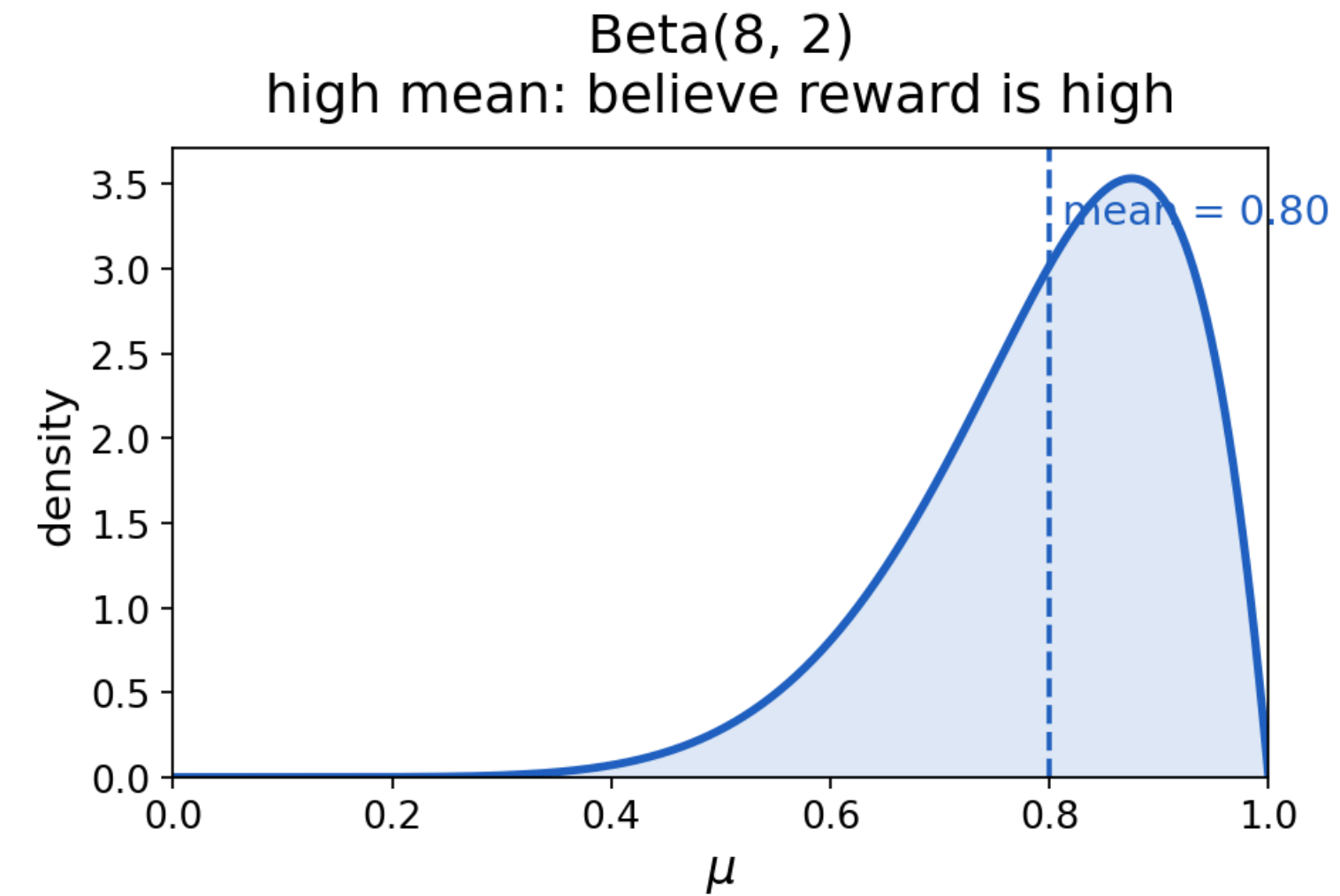
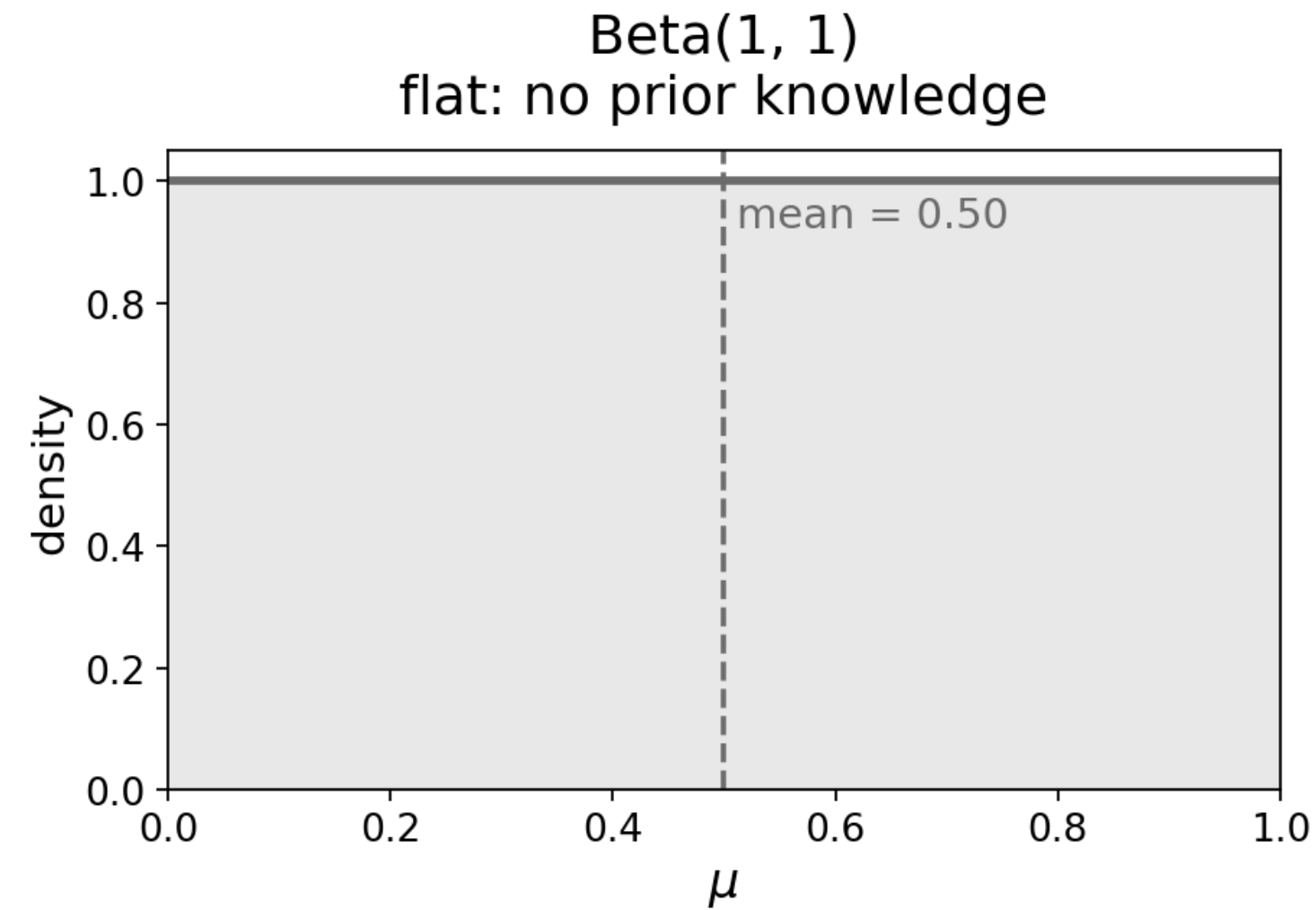
Where the density function is

$$f_{\alpha,\beta}(\mu) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)} \quad \forall \mu \in [0,1]$$

$$\tilde{\mu} \sim f_{\alpha,\beta}(\mu) \Rightarrow \mathbb{E}[\tilde{\mu}] = \frac{\alpha}{\alpha + \beta}, \quad \mathbb{V}[\mu] = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}$$

The Beta Distribution

Examples



Beta-Bernoulli Conjugacy

Updates become simplified!

For one pull of one arm:

$$X \in \{0,1\}, \quad \mathbb{P}(X = 1 \mid \mu) = \mu, \quad \mathbb{P}(X = 0 \mid \mu) = 1 - \mu$$

$\Rightarrow$ Likelihood for one data point (x)

$$\mathbb{P}(X = x \mid \mu) = \mu^x (1 - \mu)^{1-x}$$

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Prior Beta distribution:

$$\mathbb{P}(\mu) \propto \mu^{\alpha-1} (1 - \mu)^{\beta-1}$$

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Multiplying the two, we get that the posterior distribution is:

$$\mathbb{P}(\mu \mid x) \propto \mu^{\alpha+x-1} (1 - \mu)^{\beta+(1-x)-1}$$

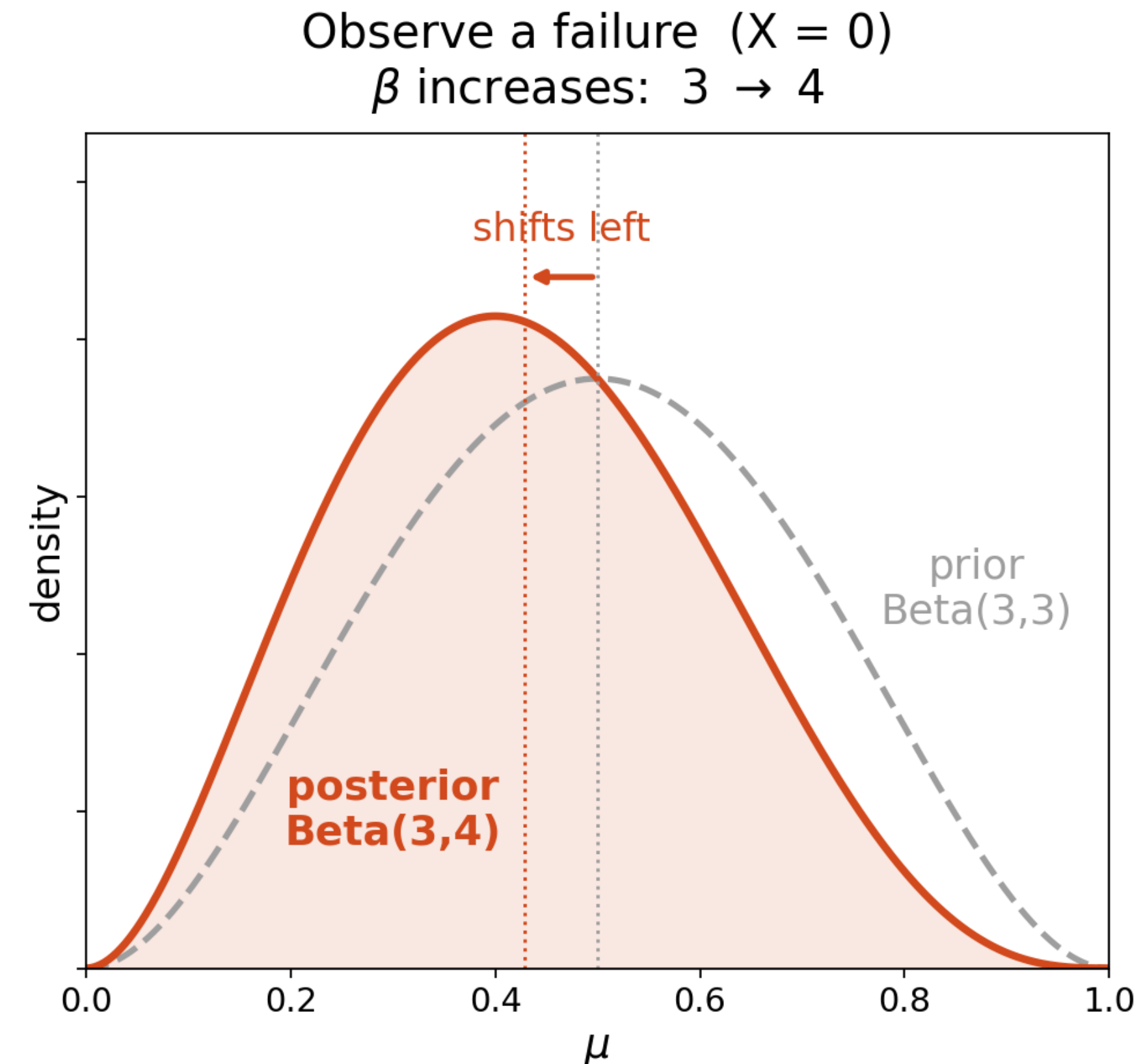
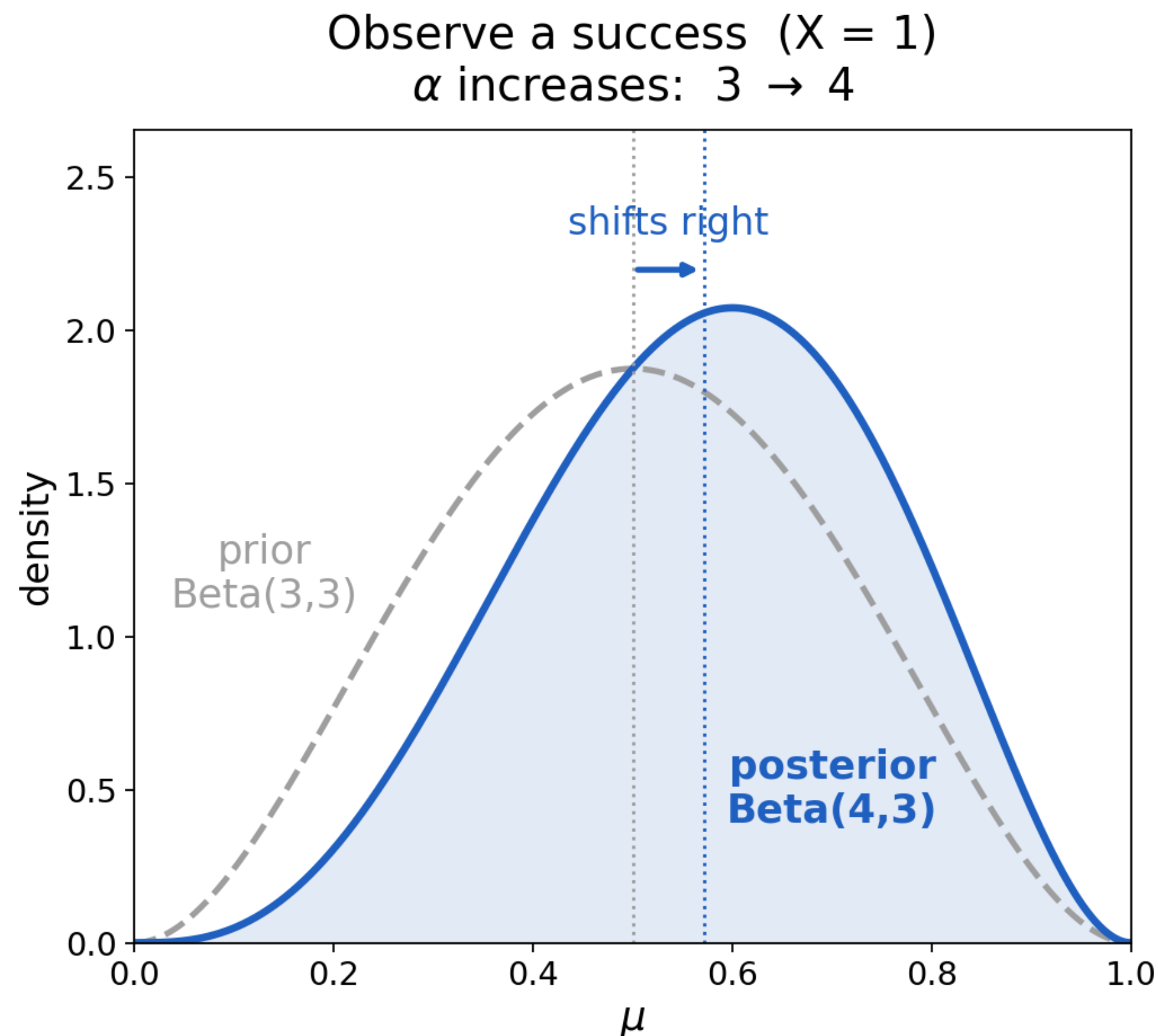
Posterior Update Illustration

Interpretation:

- If ($X = 1$): larger μ becomes more plausible
- If ($X = 0$): smaller μ becomes more plausible

$$X = 0 : \text{Beta}(\alpha, \beta) \rightarrow \text{Beta}(\alpha, \beta + 1)$$

$$X = 1 : \text{Beta}(\alpha, \beta) \rightarrow \text{Beta}(\alpha + 1, \beta)$$



Thompson Sampling

Pseudocode

Initialise: $\alpha_i = \beta_i = 1$ for all arms i

For $t = 1, 2, \dots, T$

1. For each arm i :

Sample $\tilde{\mu}_i \sim \text{Beta}(\alpha_i, \beta_i)$

2. Pull arm:

$$A_t = \arg \max_{i \in \{1, 2, \dots, K\}} \tilde{\mu}_i (t - 1)$$

3. Observe reward $X_t \in \{0, 1\}$ and update:

if $X_t = 1$: $\alpha_{A_t}++$, else: $\beta_{A_t}++$

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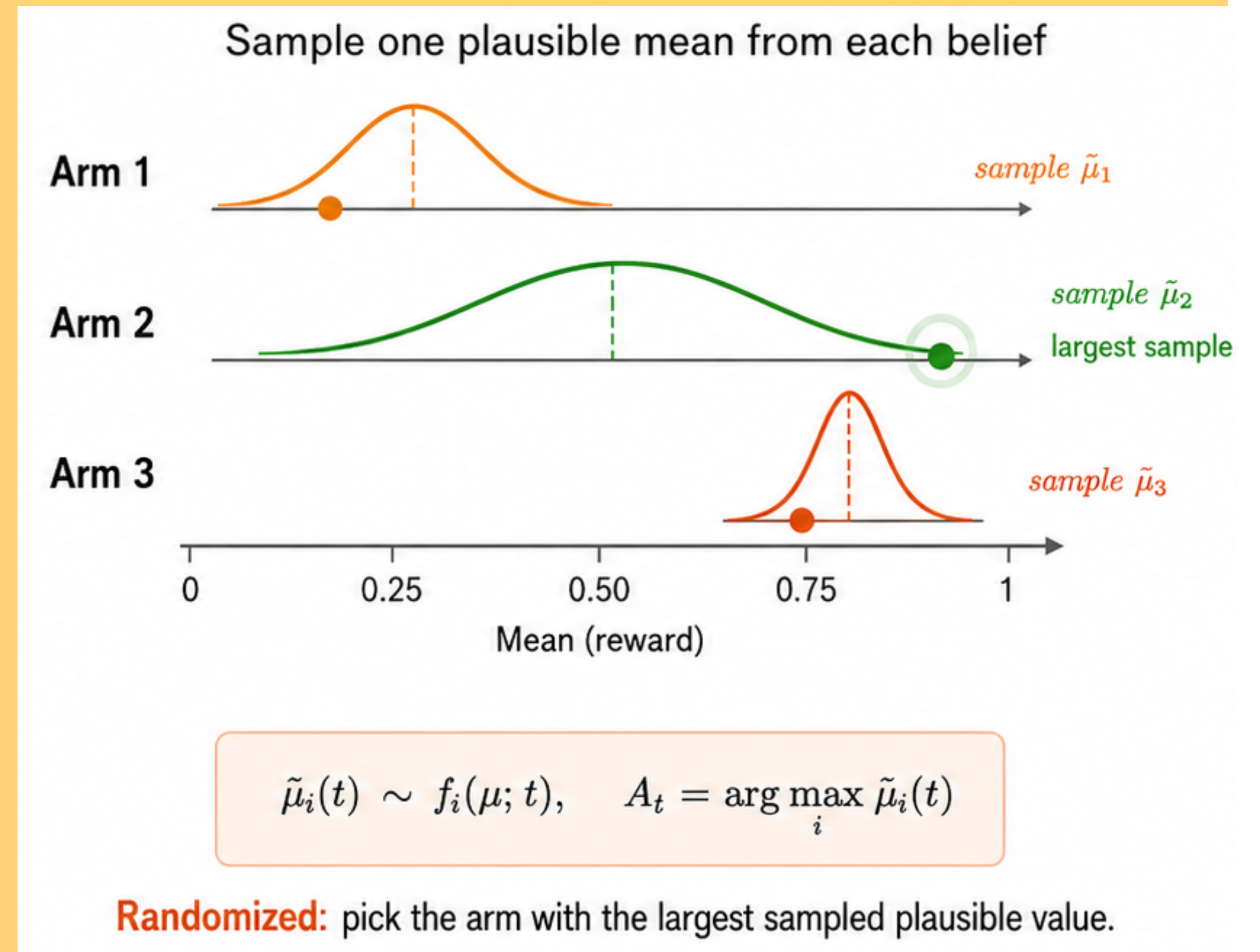
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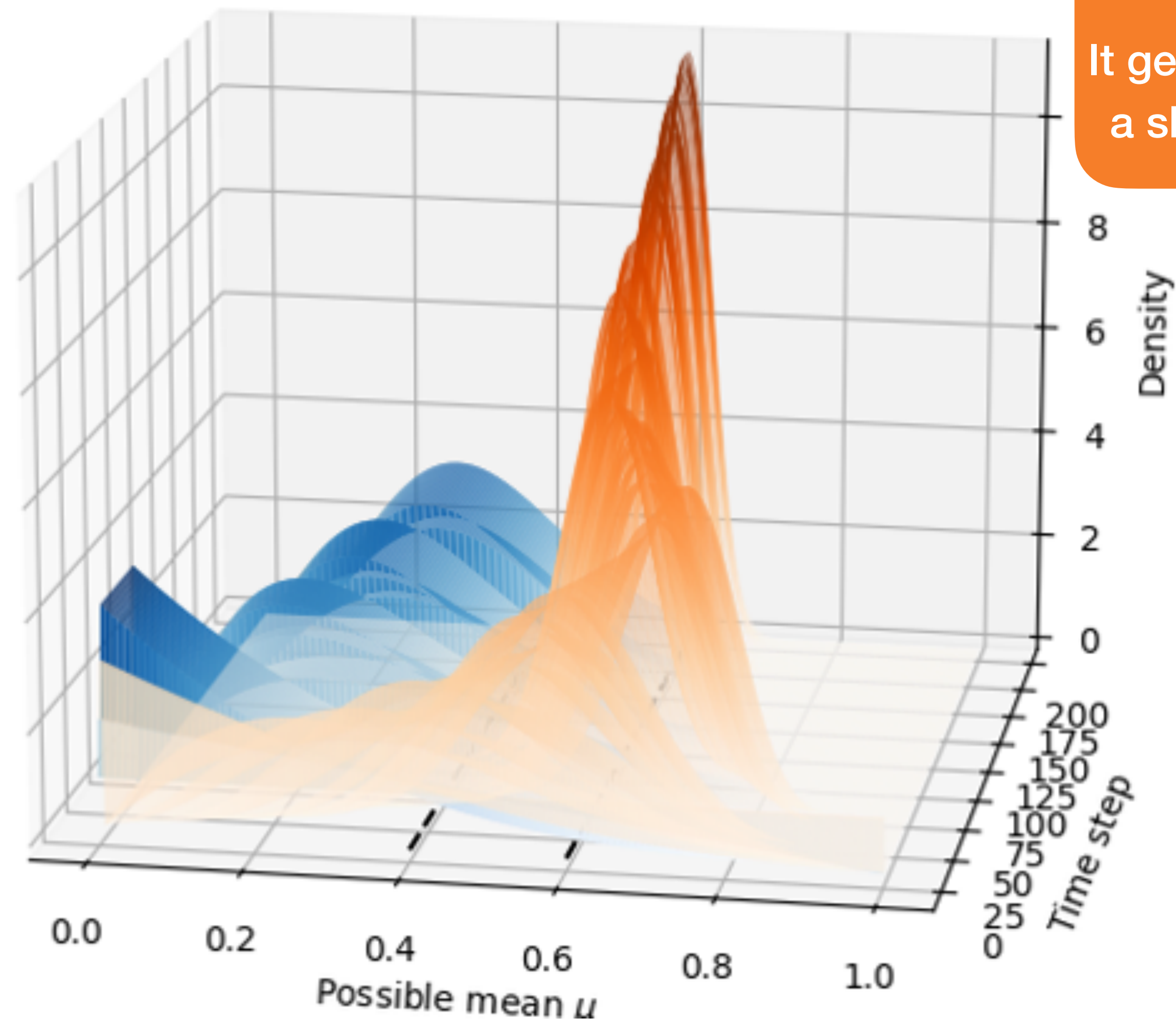
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Empirical Results

Evolution of Posteriors

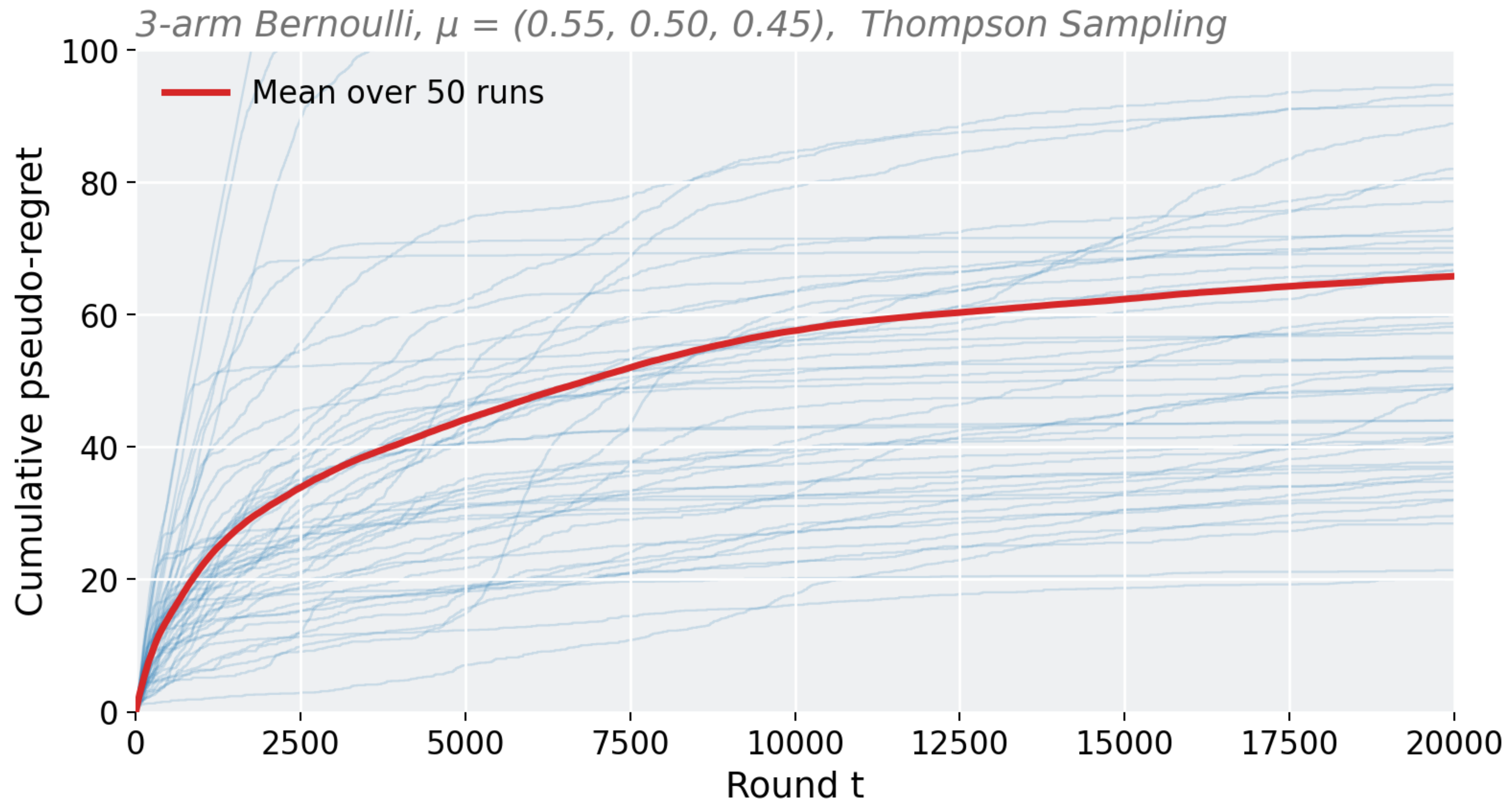
Arm 2 has true mean $\mu_2 = 0.4$
It gets pulled less often, so it has a wider posterior at $T = 200$



Arm 1 has true mean $\mu_1 = 0.4$
It gets pulled less often, so it has a sharper posterior at $T = 200$

Empirical Performance

Cumulative regret curves

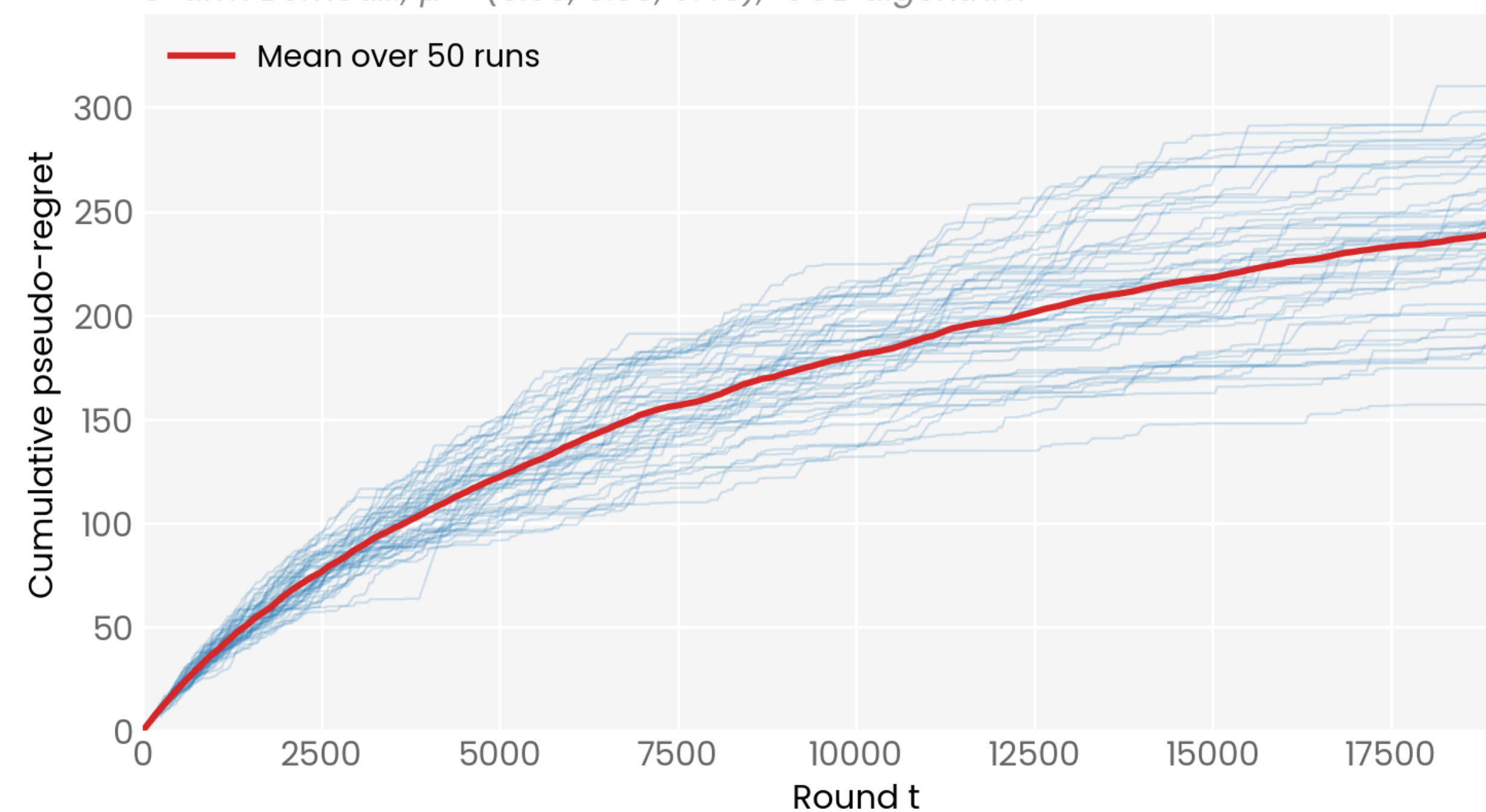


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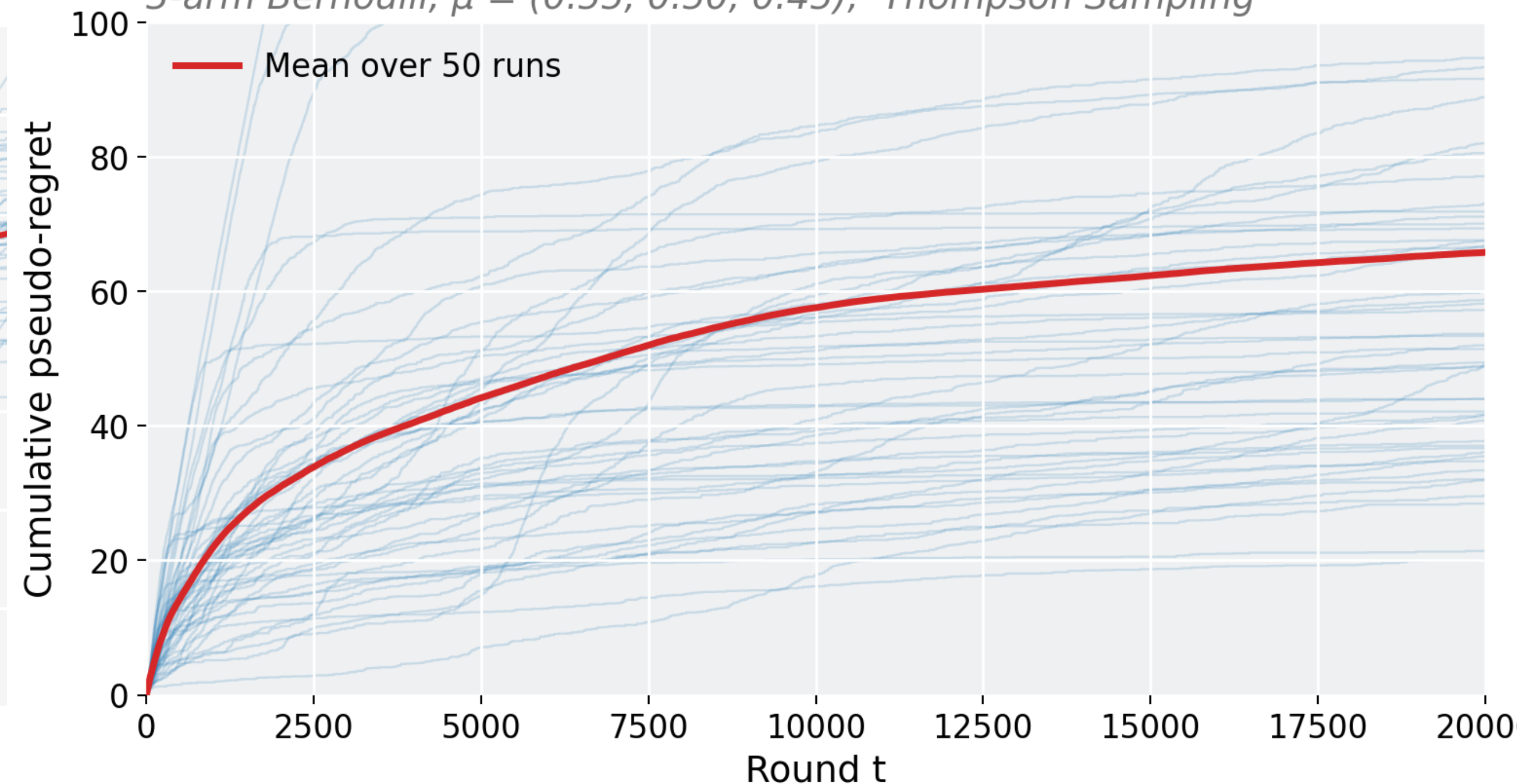
UCB

3-arm Bernoulli, $\mu = (0.55, 0.50, 0.45)$, UCB algorithm



Thompson Sampling

3-arm Bernoulli, $\mu = (0.55, 0.50, 0.45)$, Thompson Sampling

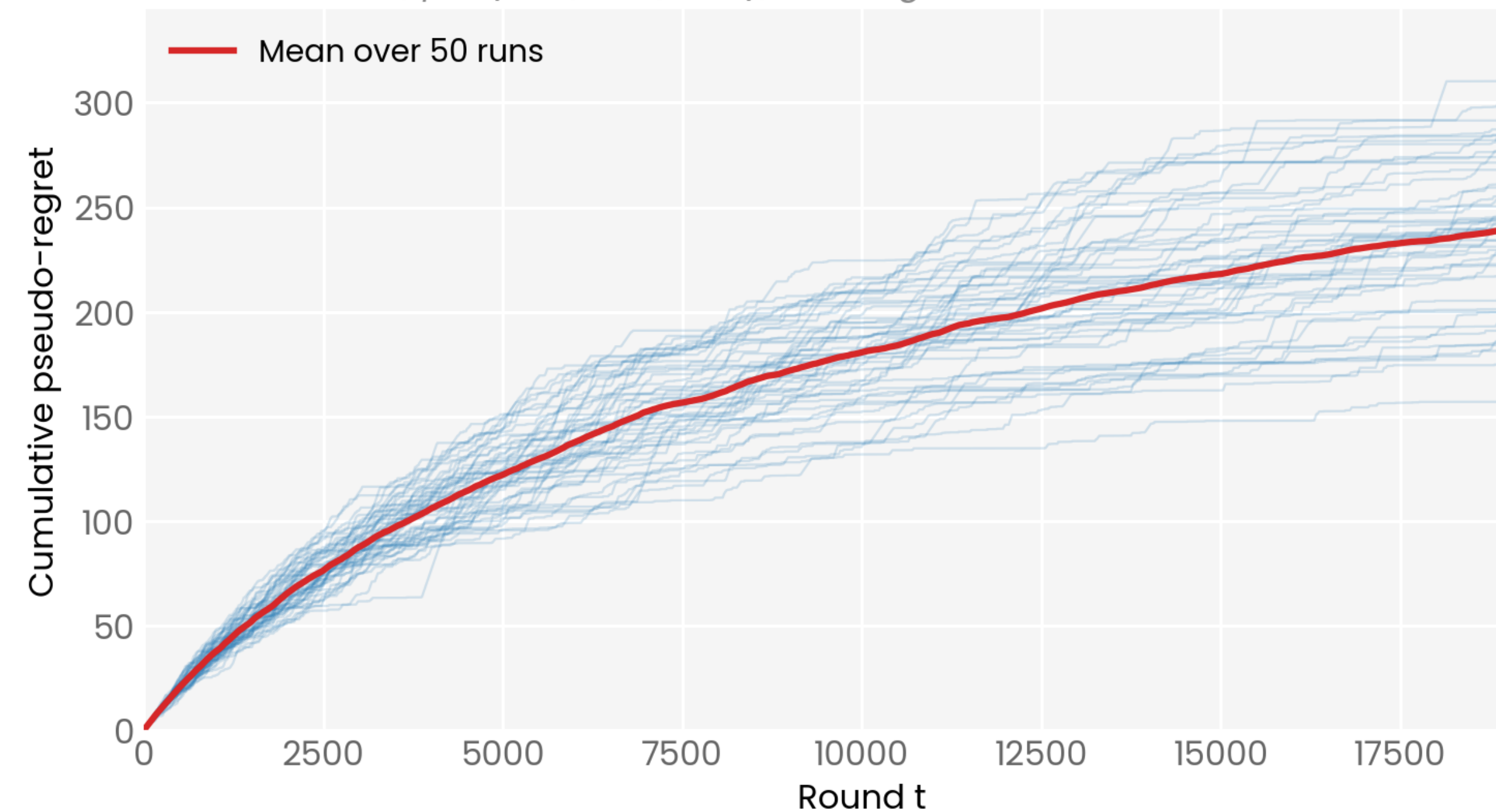


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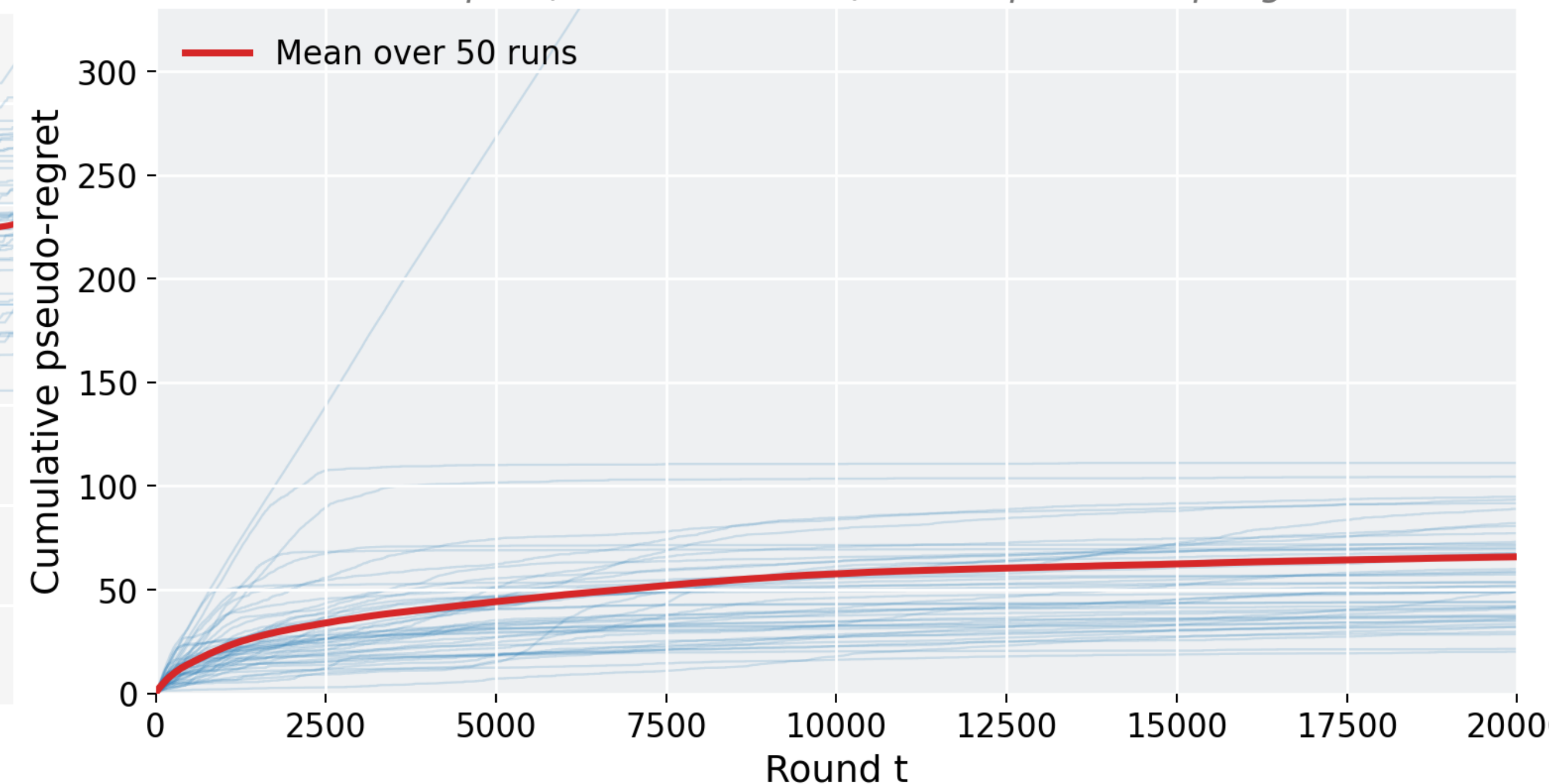
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Thompson Sampling

3-arm Bernoulli, $\mu = (0.55, 0.50, 0.45)$, Thompson Sampling



Conclusion

Thompson sampling in one line

Maintain beliefs, Sample a plausible world, act greedily in it

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Key Ingredients:

- Belief distribution for each arm, which captures uncertainty about μ_i
- Bayes' rule, used to update beliefs after each observed reward
- Posterior sampling selects arms that could plausibly be best

Conclusion

Thompson sampling in one line

Maintain beliefs, Sample a plausible world, act greedily in it

Why it works:

- Uncertain arms are explored naturally
- Confidently bad arms are rarely sampled as best
- As data grows, beliefs concentrate and exploration fades

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UCB = deterministic optimism Thompson Sampling = randomised optimism